Traffic Solutions

Traffic controllers, vehicle and pedestrian signal heads, poles, detection equipment, and backup power systems.

2.1. Traffic Signal Controllers

We offer cutting-edge traffic signal controllers. Our flagship controller is the ITC 3 Global, an adaptive traffic signal controller developed in partnership with SWARCO Technology, Denmark. We also offer other new-generation controllers that can integrate systems such as traffic counting, video surveillance, red-light violation detection, and data collection.

2.1.1. ITC 3 Controller

This controller is supplied and serviced by Global Suntech in partnership with **Swarco Technology, Denmark. It is a** versatile, high-capacity traffic controller designed for complex intersections and integration with adaptive smart-city systems. Operating on a powerful Linux ARM-based platform with dual safety processors (EN61508 SIL 3), it delivers unmatched reliability and fail-safe operation.



A versatile traffic controller that is built for the future. The controller can manage all existing signal heads and is designed to control traffic with maximum safety under all conditions.

The integrated, easy-to-use interface gives engineers a flexible solution for configuration and confirming signal status.

KEY FEATURES

Cost-effective rack system – up to 24 signal groups in a single rack

VGA Touchscreen 800x480 – Easy to maintain and program.

Innovative Adaptive Traffic Algorithms – calculate flow to capacity for every cycle and optimize cycle time and green split for a single intersection.

ITC-PC Programming tool – a powerful 3D graphical programming tool that makes it easy to configure the controller, even with advanced adaptive programs.

Network & Compatibility: Seamlessly interfaces with all major global traffic management protocols, including SCATS,

2.1.2. SW E100 controller

This controller is supplied and serviced by Global Suntech in partnership with Sinowatcher, Shenzhen, China. It is an adaptive traffic signal controller; a high-performance traffic signal control device designed for urban Intelligent Transportation Systems (ITS).

TECHNICAL FEATURES





NAME	SW E100
Signal output	56-way, extendable to 112-way
Input detection interface	28 vehicles; 8 pedestrians
Time-based schedule & period	40/24
Plans	32
Standard EIA level RS232	3, Baud rate: 1200bps-115200bps
Network interface	1, 10M/100M adaptive
USB 2.0 interface	1
GPS interface	1, with GPS time service
Pedestrian request interface	2, Dry contact signal
Flashing yellow controller interface	1 For the optocupler output, 1 is connected to OC transistor output

KEY FEATURES

It integrates adaptive, coordinated, and actuated control with multi-mode operation to dynamically optimize signal timing based on real-time traffic conditions, significantly improving roadway efficiency and traffic safety.

The controller supports time-of-day scheduling, actuated/semi-actuated control, isolated adaptive control, and offline arterial and area coordination, and can be centrally managed through a traffic control center.

It also supports optional BRT transit signal priority, VIP and special-purpose vehicle priority, and pedestrian call control, making it well-suited for complex urban road networks.

2.1.3. SW 200 Controller

This controller is supplied and serviced by Global Suntech in partnership with Sinowatcher, Shenzhen, China. It is an adaptive traffic signal controller, a high-performance traffic signal control device designed for urban Intelligent Transportation Systems (ITS).







TECHNICAL FEATURES

NAME	SW E100
Signal output	56-way, extendable to 112-way
Input detection interface	28 vehicles; 8 pedestrians
Time-based schedule & period	40/24
Plans	32
Standard EIA level RS232	3, Baud rate: 1200bps-115200bps
Network interface	1, 10M/100M adaptive
USB 2.0 interface	1
GPS interface	1, with GPS time service
Pedestrian request interface	2, Dry contact signal
Flashing yellow controller interface	1 For the optocupler output, 1 is connected to OC transistor output

KEY FEATURES

- It Powerful computing and communication with advanced ARM 32-bit chip.
- Higher running stability with SCR control on traffic signal outputs.
- Easy extension and compatibility with open NTCIP communication protocol.
- Accurate timing with GPS.
- Networking for remote control and monitoring.
- Multi working modes – fixed time, pedestrian request, green wave, actuated and manual control.
- Wiring terminals for both main and spare power supply for easy connection with electric generator during power failure.

2.1.4. Automotor Controller

This is robust and perfectly suited to the harsh operating environment of Southern Africa. It is a microprocessor-based system that integrates a modular hardware design with stage-based software to produce a highly configurable and user-friendly traffic light controller.





TECHNICAL FEATURES

NAME	Automotor
Maximum phases	16
Output	230 V, 2A fused solid-state
Operating voltage	190 V ac – 270 V AC
Plans	32
Bad return delay	0.4sec
Streams	2
Standard	SABS standard
Modes of operation	Fixed, vehicle actuated, Manual flash, fault, emergency, flashing yellow.
USB 2.0 interface	1
GPS interface	1, with GPS time service
Pedestrian request interface	2, Dry contact signal

KEY FEATURES

It has a simple, robust user interface that provides easy access to information such as synchronization status, the current plan and stage, RTC setup, and fault-finding diagnostics.

It can be configured as a 4-, 8-, 12-, or 16-phase traffic light controller, with or without detection. All controller cabinets are weatherproof and specially designed to deter vandals.

The cabinets are made from 2 mm hot-dip galvanized steel for strength and can be powder-coated as an optional extra for aesthetics and added corrosion resistance.

2.1.5. Synthel controller

This is robust and perfectly suited to the harsh operating environment of Southern Africa. It offers all the sophistication required to address today's traffic engineering challenges while providing a non-mental solution. The system is fully modular and robust, but above all, it is simple.



TECHNICAL FEATURES

NAME	Automotor
Maximum phases, stage	8 / 12
Output	230 V, 2A fused solid-state
Stream	2
Operating voltage	190 V ac – 270 V AC
Plans	32
Bad return delay	0.4s
Standard	SABS standard
Modes of operation	Fixed, vehicle actuated, Manual flash, fault, emergency, flashing yellow.
USB 2.0 interface	1
GPS interface	1, with GPS time service
Pedestrian request interface	2, Dry contact signal

KEY FEATURES

It provides comprehensive feedback on module status and can be easily detected, enabling support staff to quickly and effectively identify faults without special tools or support device.

It it retains fault conditions during outages.

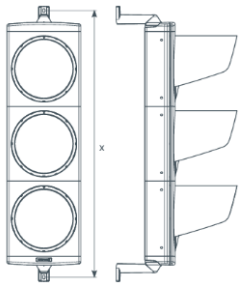
The cabinets are made from 2 mm hot-dip galvanized steel for strength and can be powder-coated as an optional extra for aesthetics and added corrosion resistance.

2.2. LED Traffic Light

We supply energy-efficient LED traffic signal heads in 100 mm, 210 mm, and 300 mm sizes, with vehicle, pedestrian, and directional arrow displays. UV-stabilised polycarbonate housings provide weather resistance. Our range also includes flashing warning lights, road studs, and radar speed signs.

2.2.1. Futura/CIWAY LED signal heads

Durable, energy-efficient LED signal heads for traffic signals, tollgates, and booths. Available with a yellow module or integrated tri-color countdown timer, they feature UV-stabilised polycarbonate housings and wide-opening access for easy assembly and maintenance.



HEIGHT X

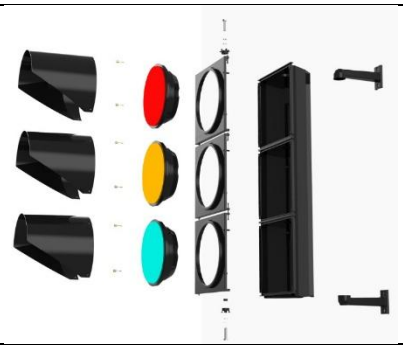
100 mm	1 aspect	2 aspects	3 aspects	4 aspects
PC 70	309 +/- 14	444 +/- 14	579 +/- 13	714 +/- 13
200 mm	**1 aspect**	**2 aspects**	**3 aspects**	**4 aspects**
PC 70	440 +/- 13	700 +/- 12	960 +/- 11	-
PC 175	456 +/- 7	716 +/- 6	976 +/- 6	-
Alu 105 / 183 / 240	447 +/- 16	707 +/- 15	967 +/- 15	1227 +/- 15
300 mm	**1 aspect**	**2 aspects**	**3 aspects**	**4 aspects**
PC 70	530 +/- 13	880 +/- 12	1230 +/- 11	-
PC 175	546 +/- 7	896 +/- 6	1246 +/- 6	-
Alu 105 / 183 / 240	537 +/- 16	887 +/- 15	1237 +/- 15	1587 +/- 15



Technical details	
Material	UV-stabilised polycarbonate
Diameter	100/200/300 RYG
IP protection	IP55
Impact resistance	IR3
Temperature	-40 C to +60 C
Protection class	SK II
LED-optic	100/200/300 mm LED modules
Certificates	EN12368, CE
Housing color	Black.

2.2.2. ST 320-Clear Lens Full ball

These are reliable vehicle traffic signals designed for modern road infrastructure and smart city traffic systems. Built with a rugged aluminum alloy housing, high-efficiency SMD LED light source, and UV-resistant optical lenses, they deliver stable performance, strong visibility, and long service life in demanding outdoor environments. Its modular design and wide voltage compatibility ensure easy installation, low maintenance, and dependable operation across traffic control applications.



HEIGHT X

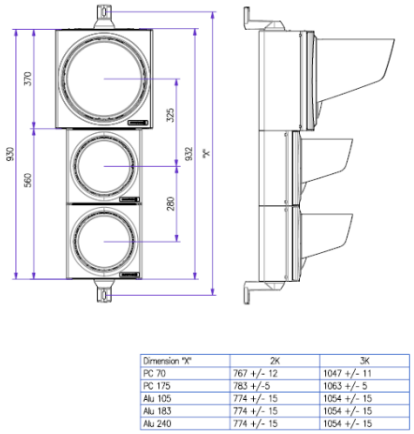
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PC 175	546 +/- 7	896 +/- 6	1246 +/- 6	-
Alu 105 / 183 / 240	537 +/- 16	887 +/- 15	1237 +/- 15	1587 +/- 15



Technical details	
Material	UV-stabilised polycarbonate
Diameter	200/300| RYG
IP protection	IP65
Voltage	AC85-265Vac
Temperature	-40 C to +60 C
Protection class	SK II
LED-optic	High-intensity SMD with optimized optical visibility.
Certificates	EN12368, CE RoHS
Housing color	Black

2.2.3. FCiway 300+200 RYG.

Durable, energy-efficient LED signal heads for traffic signals in RYG, also available with a yellow module or an integrated tri-color countdown timer. They feature UV-stabilised polycarbonate housings and wide-opening access for easy assembly and maintenance.



Dimension "X"	2X	3X
PC 70	767 +/- 12	1047 +/- 11
PC 175	783 +/-5	1063 +/- 5
Alu 105	774 +/- 15	1054 +/- 15
Alu 183	774 +/- 15	1054 +/- 15
Alu 240	774 +/- 15	1054 +/- 15

Features and Benefits:
- Wide operating voltage and transparent polycarbonate optical lens, impact-resistant.
- Anti-phantom effect with optimal viewing angle
- High-intensity LED light source-low energy consumption
- 5mm LED (DIP) by EPISTAR
- Long life cycle-more than 80,000 working hours
- Highly durable polycarbonate housing, UV-light resistant
- Modular design-easy installation and maintenance
- Easy access using a wingnut-open system
- Fixation system with L-shaped/arm-shaped mounting brackets
- EN12368 Certified, CE & RoHS Compliant, ISO9001 Certified Manufacturing



Technical details	
Material	UV-stabilised polycarbonate
Diameter	200/300| RYG
IP protection	IP65
Voltage	AC85-265Vac
Temperature	-40 C to +60 C
Protection class	SK II
LED-optic	High-intensity SMD with optimized optical visibility.
Certificates	EN12368, CE RoHS
Housing color	Black

2.2.4. Pedestrian Signal RG

Pedestrian traffic lights indicate when pedestrians can safely cross the road. They reduce conflicts between pedestrians and vehicles, lower accident risk, and maintain smooth traffic flow at intersections, supporting all road users.



Features and Benefits:
- Wide operating voltage and transparent polycarbonate optical lens, impact-resistant.
- Anti-phantom effect with optimal viewing angle
- High-intensity LED light source-low energy consumption
- 5mm LED (DIP) by EPISTAR
- Long life cycle-more than 80,000 working hours
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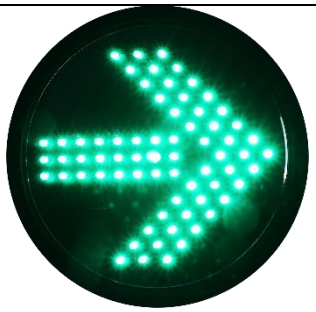
Technical details	
Material	UV-stabilised polycarbonate
Diameter	200/300| RG
IP protection	IP65
Voltage	AC85-265Vac
Temperature	-40 C to +60 C
Protection class	SK II
LED-optic	High-intensity SMD with optimized optical visibility.
Certificates	EN12368, CE RoHS
Housing color	Black

2.2.5. Accessories

Our traffic light countdown timers display the remaining time of the current signal phase, helping drivers and pedestrians anticipate signal changes. Designed for a range of traffic applications, they provide clear timing information to support safer, more orderly road use.

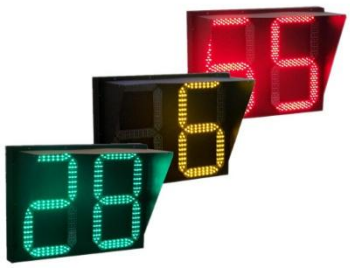


Features and Benefits:
- Wide operating voltage and transparent polycarbonate optical lens, impact-resistant.
- Anti-phantom effect with optimal viewing angle
- High-intensity LED light source-low energy consumption
- 5mm LED (DIP) by EPISTAR
- Long life cycle-more than 80,000 working hours
- Highly durable polycarbonate housing, UV-light resistant
- Modular design-easy installation and maintenance
- Easy access using a wingnut-open system
- Fixation system with L-shaped/arm-shaped mounting brackets
- EN12368 Certified, CE & RoHS Compliant, ISO9001 Certified Manufacturing

		
	Technical details	
	Material	UV-stabilised polycarbonate
	Diameter	200/300| RG
	IP protection	IP65
	Voltage	AC85-265Vac
	Temperature	-40 C to +60 C
	Protection class	SK II
	LED-optic	High-intensity SMD with optimized optical visibility.
	Certificates	EN12368, CE RoHS
	Housing color	Black

2.2.6. Countdown timer

Our traffic light countdown timers display the remaining time of the current signal phase, helping drivers and pedestrians anticipate signal changes. Designed for a range of traffic applications, they provide clear timing information to support safer, more orderly road use.

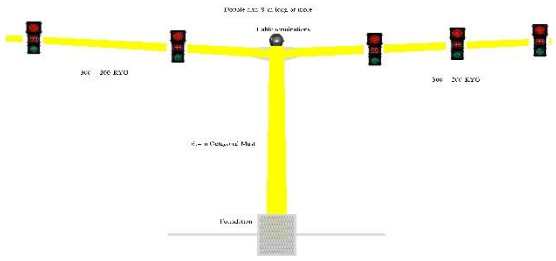
	**Features and Benefits:** - Wide operating voltage and transparent polycarbonate optical lens, impact-resistant. - Anti-phantom effect with optimal viewing angle - High-intensity LED light source-low energy consumption - 5mm LED (DIP) by EPISTAR - Long life cycle-more than 80,000 working hours - Highly durable polycarbonate housing, UV-light resistant - Modular design-easy installation and maintenance - Easy access using a wingnut-open system - Fixation system with L-shaped/arm-shaped mounting brackets - EN12368 Certified, CE & RoHS Compliant, ISO9001 Certified Manufacturing
	
	Technical details
	Material | UV-stabilised polycarbonate
	Diameter | 200/300| RG
	IP protection | IP65
	Voltage | AC85-265Vac
	Temperature | -40 C to +60 C
	Protection class | SK II
	LED-optic | High-intensity SMD with optimized optical visibility.
	Certificates | EN12368, CE RoHS
	Housing color | Black

2.3. Traffic Poles

The poles are designed using computer software to analyze the pole shaft, cantilever arm, base plate, and anchor bolts. All materials used to fabricate the pole components will comply with international standards and project specifications. The anchor bolts, including nuts and washers, are hot-dip galvanized. Traffic signals are attached to the cantilever arms at specified locations based on the road layout.

2.3.1. Overhead/Cantilever Poles

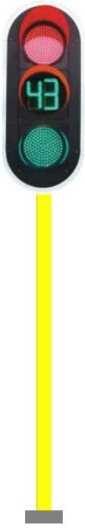
The vertical is 6.4 m, either the same section or tapered. It carries a cantilever arm fixed at the top with thick flanges, secured with bolts and nuts. The arm length depends on road design and traffic requirements, and it is installed at a 5-degree angle. It supports at least three signal heads and one countdown timer while providing sufficient clearance for abnormal loads.

<table>
<tr>
<td>  </td>
<td>Features and Benefits:
- Hot galvanizing or process coating ; Powder colors optional :silver grey ; milky white ; black ; dark green ; etc.According to RAL code.
- Customizable poles, and clearly visible traffic signal poles support organized pedestrian movement and protect vulnerable road users near schools and crowded public areas.
- Factories, logistics hubs, and gated communities integrate signal poles to control vehicle access and guide traffic.
- Design according to engineering drawings.</td>
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<table>
<tr><td colspan="2">Technical details</td></tr>
<tr><td>Material</td><td>Mild steel</td></tr>
<tr><td>Diameter</td><td>350/300/250 mm</td></tr>
<tr><td>Final finish</td><td>Automotive yellow</td></tr>
<tr><td>Corrosion resistance</td><td>Hot-dip galvanized/ equi</td></tr>
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The actual drawing is provided as an annexure; the figure shows a snapshot of the engineering drawing for the two designs. Importantly, the pole is fixed to the ground using anchor bolts, nuts, and washers, which are hot-dip galvanized or equivalent. The signal heads are then mounted on the cantilever arm at predetermined locations in accordance with the road layout. The heads can be clamped in different ways.

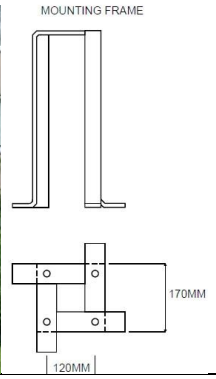
2.3.2. Standard Poles

In addition, the design incorporates standard poles that are 4,000 mm high, made of mild steel tubing, and either tubular or square, with an outer diameter of 114 mm and a thickness of 2 mm. In some cases, we increase the pole thickness to 3.9 mm to reduce damage or impact from accidents. However, we normally protect the poles with barriers. In both cases, cable terminations are made at the top using the pole terminal box. Our design includes a foundation frame and a flange on the pole, making installation easy.



Features and Benefits:
- Hot galvanizing or process coating ; Powder colors optional :silver grey ; milky white ; black ; dark green ; etc.According to RAL code.
- Customizable poles, and clearly visible traffic signal poles support organized pedestrian movement and protect vulnerable road users near schools and crowded public areas.
- Factories, logistics hubs, and gated communities integrate signal poles to control vehicle access and guide traffic.





Technical details	
Material	Mild steel
Diameter	114 mm outside
Final finish	Automotive yellow
Mounting	Brackets

1. **LED Lighting:** Streetlights, floodlights, and luminaires for high-mast and area lighting applications.
2. **Engineering Consulting:** Presented as services rather than products.

Product availability, specifications, and warranties are confirmed for each quotation

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